Optimal Price Discovery Through Sequential Experimentation

Aeden Rask, Diego McGrath, Imaan Shakoor, Simeng Zhang, Enya Tran, Megan Timmes

This lab addresses the challenge of optimal price discovery through controlled experimentation in a demand setting where the relationship between price and consumer purchase probability is unknown. We conducted a series of A/B tests across a grid of candidate prices between $0 and $70, collecting sales data from customers with arrival counts drawn from a Poisson distribution to estimate the expected revenue per customer at each price point. The goal was to algorithmically identify the globally optimal price that maximizes per-customer profit, while managing the inherent trade-off between exploration (gathering sufficient data to reduce uncertainty at each price) and exploitation (allocating more observations to promising price points). We compared our results to the benchmark of $11.925 expected profit per customer at the true optimal price.

Due to the fact that each customer's purchase is random, and the number of arrivals varies from day to day, a difference in observed revenue between two prices could be noise rather than a real change in demand. Therefore, we treated every price as an experiment and never compared prices by their raw averages alone. For each price, we stored customer level revenue (price x purchase indicator) and computed the mean revenue per arriving customer, its standard error, and a 90% confidence interval using the normal critical value 1.645. To compare prices, we used a two-sample Z test, where we took the difference between each candidate's mean and the current leader's mean, with standard error. If the 90% interval for that difference was entirely below zero, that candidate was evidently worse and was dropped. If the interval reached zero or higher, we kept it and retested it. Furthermore, we chose 90% for our interval as a wider one would keep too many weak prices alive and cost more customers. Because failing to exclude zero can reflect low power rather than a true tie, we did not treat an overlapping interval as evidence that two prices were equal. Instead, we gave those prices more data, and we pooled all batches collected at a price, so every added batch shrank its standard error and made later comparisons better.

Price selection combined brute-force grid search with sequential refinement, where Round 1 tested a coarse $5 grid from $5 to $70 with about 200 customers each, which prevented us from being stuck at one of the local peaks in the demand curve. We found that $30 had the highest mean, with local maxima on the grid around $10, $30, $40, and $50. We placed two new interior prices in each region and assigned 200 customers to prices near the current leader, $30, and 100 customers to prices in other regions, since prices near the leader are the most likely candidates for the optimal price and require more data to distinguish from it. From this, we found a new best price of $28.33. With this in mind for later rounds, we reallocated customers using the difference confidence intervals, where prices not worse than the best price ($11.67, $25, $28.33, $30) received 200 more customers each. In round 4, $11.67 was dropped due to its negative interval, and we then added midpoints between $28.33 and its neighbors, $26.66 and $29.16, to narrow the search to a finer step size without creating a full new grid. The final price was chosen

based on having the highest mean revenue per customer across all pooled data, which was $28.33 at about $12.88 per customer. Finally, as a visual check, Figure 1 shows the 90% confidence intervals along with the mean revenue per customer at each tested price, where $28.33 has the highest observed mean.

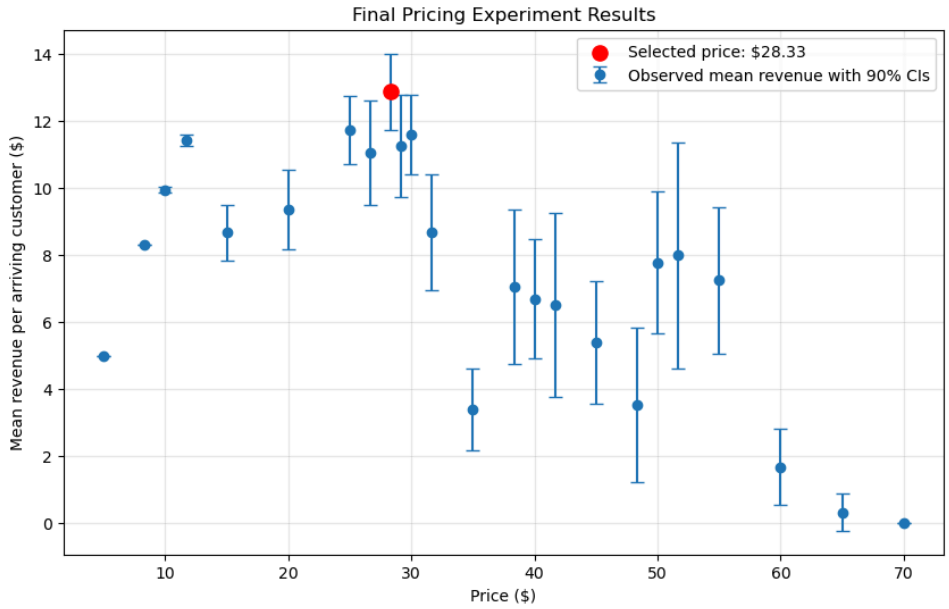


Figure 1: Price vs Mean Revenue per Arriving Customer, Final Round Results

The biggest problem we ran into early on was that the demand curve had more than one peak, and with only about 200 customers per price, it was hard to tell which peaks were real. Our code flagged $10, $30, $40 and $50 as possible peaks, but at least one of these was probably just noise: $35 only made $3.41 per customer, so $40 looked like a peak mostly because the price next to it came in so low. Revenue also fell off a cliff between $30 and $35($12.29 down to $3.41), so if we had only searched around one starting price, we could have gotten stuck on the wrong peak. Instead of trusting the highest average, we tested new prices in every candidate region in round 2, giving the far-away regions only about 100 customers each to save costs, and then used the difference confidence intervals to decide which prices to keep. One thing that surprised us was that $11.67 made it into round 3 even though its mean was lower than $28.33. Nearly everyone buys at $11.67, so its own standard error was tiny, but $28.33 had a much bigger standard error (about $1.03). That made the difference interval wide enough to reach zero, so we failed to reject that the two prices were equal. Once we pooled in the round 3 customers, the standard error at $28.33 shrank to $0.689, the interval for $11.67 fell completely below zero, and we dropped it.

The second challenge was that learning about prices costs money. Our first grid used 2,782 customers, which is more than half of the 4,958 we experimented on, and it only earned about $6.43 per customer. A lot of that came from the 1,567 customers who saw prices of $35 or higher and only brought in about $4.02 each. Round 1 alone accounts for about $15,280 of the

$18,433 gap between our revenue and the optimal revenue, which is the main reason we only captured 68.82%. We cut costs in later rounds by dropping prices that were clearly worse and by only testing the midpoints right next to the leader instead of building a whole new grid. We also noticed that $28.33 first showed $13.04 per customer, but after 231 more customers its pooled mean dropped to $12.88. That's still above the $11.925 the best possible price is expected to earn, so the leader was probably helped by a lucky sample, which is why we retested it before picking it instead of trusting one batch. If we did this again, we would still cover the full $0 to $70 range in round 1 so we do not miss the global peak, but we would use fewer customers per price there, since most of those prices were clearly bad after one look.

The price optimization was conducted in 4 rounds. Each round provided more information about the underlying demand curve and narrowed in on the global maximum. For round 1, a coarse grid, 14 prices were tested at about 200 customers each. The results from this grid showed a best observed price of $30 that was statistically significant at a 90% confidence interval from the other peaks. The results of the next round, a grid of smaller regions, tested 200 customers for prices near the current best ($30) and around 100 customers for further prices. After this process, the new leader was $28.33. The next round was focused on building confidence intervals for the differences of every tested price and the current leader. The prices that had a difference CI reaching 0 or higher were retained for retesting. At this point the remaining prices were $11.67, $25, $28.33, and $30. To find the final answer, these prices were tested on 200 additional customers. For the final round, $11.67 had been shown as clearly worse from the difference CIs and had been dropped. Refining continued with added price points at midpoints between $25 and $30 to assist in finding the optimum point. At the end of this process $28.33 had the highest observed mean over all the data, leading to this being the price picked. The total customers tracked over the experimentations was 4,958. The optimum profit for this would have been 11.925 * 4958, which is $59,124.15. During our experiments on 4,958 customers, the revenue earned was $40,690.70. This is 68.82% of the optimum revenue. This gap is a result of the experimentation where the grids at the start tested low revenue prices.